

Taledge AI Intelligence: Core Architecture & Scoring

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This document outlines the ultra-complex, world-class algorithmic foundation of the Taledge Dual-Track Success Intelligence Platform. It details the adversarial AI interviewer methodology, cognitive load mapping, and the exact multi-layered calculation of the Fit Score and Success Probability.

1. The Interaction Flow: Continuous Evaluation Loop

The Taledge AI Interviewer is not a static Q&A bot. It operates as an **Adversarial Engineering Leader** (in Technical Mode) and an **Elite Behavioural Psychologist** (in Behavioural Mode). The interaction is continuous, dynamic, and adaptive.

1.1 Context Payload & Initialization

Before the interview begins, the frontend compiles a rigid JSON context payload derived from the candidate's parsed resume, target role, and identified academic/project history.

1.2 The Adversarial Prompt Engine

The AI is instructed strictly through dynamic role-based prompting. It possesses multilingual abilities (English, Hindi, Hinglish) and adapts instantly.

- **Cognitive Load Stress-Testing:** If a candidate answers a standard technical question perfectly, the AI instantly increases cognitive load by combining concepts or introducing edge-case system failure states to test their working memory.
- **DNLA Psychometric Probing:** In the behavioural phase, the AI actively bypasses generic STAR (Situation, Task, Action, Result) answers to probe emotional regulation, ethical boundaries, and defensive mechanism triggers.

1.3 Execution Pipeline

1. **Real-time STT:** Browser-native Webkit SpeechRecognition transcribes the candidate's speech.
2. **LLM Routing:** The transcription is passed to the Taledge API (`/api/interview/voice/route.ts`), which interfaces with OpenRouter to process the transcript through the adversarial persona.
3. **Real-time TTS:** The generated text is vocalized back to the candidate via the browser's native `SpeechSynthesis` .
4. **Proctoring Overlay:** Background listeners track `visibilitychange` events to detect window-switching, applying immediate cheating penalties and logging violation counts.

2. The Fit Score & Success Probability Calculation

The ultimate output of the platform is the **Success Potential Score (0-100%)**. This is calculated mathematically across 5 distinct feature matrices evaluated by the LLM post-interview.

Matrix 1: Technical Interview Signals

Competency Domain	Extracted AI Metric	Description / Proxy Measure
Accuracy & Coverage	Tech Accuracy Score	Percentage of core domain questions answered correctly.
	Difficulty Weighted Accuracy	Weighted correctness (harder questions carry exponential weight).
Problem Solving Depth	Solution Correctness	Binary mapping of full vs. partial solution.
	Approach Structure	Logic decomposition and solution construction pipeline.
	Multi-Approach Capability	Did the candidate proactively propose alternative solutions?
Thinking Quality	Reasoning Clarity	NLP scoring of explanation clarity.
	Conceptual Correctness	Even if the final answer failed, were the underlying mechanisms correct?
	Error Recovery	Ability to detect and correct their own mistakes under pressure.
Coding Quality	Efficiency & Correctness	Algorithmic Big-O evaluation and clean code principles.
Micro-Expression Proxies	Response Latency	Average hesitation time before answering (proxy for confidence).
	Latency Variance	Guessing vs. active deep thinking mapping.
	Hint Dependency	Frequency of required hints for complex architectures.

Matrix 2: Resume & Profile Features

Competency Domain	Extracted AI Metric	Description / Proxy Measure
Skill Mapping	Skill Match Score	Cosine similarity between parsed resume skills and Job Description.
	Core Skill Percentage	Percentage of non-negotiable required skills present.
Project Reality Index	Project Relevance	Solving a real-world enterprise problem vs. generic academic project.
	Impact Quantification	Hard outcomes, user scale, and latency improvements over vague claims.
Academic Signals	Academic Consistency	Longitudinal grades trend.

Matrix 3: DNLA Social Competence Assessment

Imported via APIs from Germany, the DNLA report serves as the baseline psychological profile.

- **Achievement Dynamics:** Drive, Motivation, Self-confidence.
- **Interpersonal Relations:** Empathy, Assertiveness, Sociability.
- **Will to Succeed:** Systematic mentality, Initiative.
- **Stress Capacity:** Feedback reaction, Resilience under pressure, Outlook.

Matrix 4: Behavioural Interview Features

The AI conducts a targeted interview designed to validate the DNLA baseline against actual anecdotal stress-testing.

Competency Domain	Extracted AI Metric	Description / Proxy Measure
Communication	Clarity & Structure	STAR approach adherence and calibrated verbosity (too short vs. rambling).
Content Quality	Relevance & Specificity	Did the response answer the intent? Real examples vs. generalized fluff.
Ownership & Attitude	Blame vs. Accountability	"I" vs "Team" vs "Them" linguistic indexing when discussing failures.
Consistency Checks	Internal Consistency	Identifying contradictions across responses and alignment with resume claims.
Cultural Fit	Collaboration Signal	Perspective-taking and stakeholder map understanding.

Matrix 5: Cross-Component Features (The Red Flags)

The final validation layer cross-references data across all 4 previous matrices to identify fatal hiring risks.

Cross-Component Metric	Warning Condition	System Action
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Tech vs. Resume Gap	High resume claims but low technical interview performance.	RED FLAG
Confidence vs. Accuracy	High confidence linguistic markers with low actual accuracy (Overconfidence).	RED FLAG
Behaviour vs. DNLA	Claimed personality traits in DNLA do not match interview actions.	Score Penalty

3. Final Aggregation

The final Fit Score is an algorithmic synthesis of the above 5 matrices. By combining the rigid diagnostic power of German psychometrics (DNLA) with an Adversarial AI technical evaluation, the Taledge platform achieves what no other system can: **a truly holistic, deeply human, yet computationally flawless measure of success probability.**